

Benchmark Results and Evaluation Metrics

The final benchmark contains 160 financial regulation questions. It was designed as a targeted hallucination stress test after analyzing where the baseline RAG system failed. It focuses on false premises, unsupported source transfer, and precise operational details that are not actually stated in the cited evidence.

Final Benchmark Results

System Variant	Expected Behaviour Match	Answer Rate	Abstain Rate	Forbidden Claim Hit Rate
Baseline RAG	83.125%	100.000%	0.000%	0.625%
RAG + Detector	95.625%	49.375%	50.625%	0.625%
RAG + Detector + Stochastic Gate	98.125%	45.000%	55.000%	0.000%

Benchmark Category Distribution

Question Category	Count	Evaluation Purpose
cross_source_nuanced	72	Tests whether the system rejects unsupported transfer of evidence between regulators, documents, or supervisory contexts.
low_evidence_policy	40	Tests whether the system avoids inventing exact operational details when evidence only supports a broad topic.
false_premise	32	Tests whether the system rejects misleading questions that assume a fabricated regulatory requirement is true.
factual_supported	16	Checks that the system can still answer normally when the requested information is directly supported by retrieved evidence.

Metric Definitions

Metric	Meaning	Why It Matters
Expected Behaviour Match	Percentage of questions where the system performed the intended action: answer, qualify, refuse, or reject the false premise.	Main end-to-end quality metric. It evaluates behaviour, not only fluent wording.
Answer Rate	Percentage of questions where the system returned a substantive answer instead of abstaining.	Measures usefulness, but a high value can be risky if unsupported questions are answered confidently.
Abstain Rate	Percentage of questions where the system refused to answer because evidence was insufficient or risky.	In financial regulation, abstaining can be safer than giving an unsupported answer.
Forbidden Claim Hit Rate	Percentage of questions where the answer included a claim explicitly marked as fabricated or unsupported.	Direct hallucination-risk metric. Lower is better; 0.000% means all benchmark-defined forbidden claims were avoided.

Category Explanations

cross_source_nuanced

Questions contain evidence from more than one document or regulator. The risk is incorrectly transferring a rule from one authority to another.

low_evidence_policy

Evidence supports a general topic but not a precise detail. The system should qualify or abstain instead of inventing deadlines, templates, portals, or tools.

false_premise

The question assumes a fabricated regulatory requirement is true. The system should reject the premise or abstain.

factual_supported

Directly supported sanity questions. They prevent the system from appearing safe only because it refuses too often.

Key Finding

Baseline RAG answered every question, which made it useful but vulnerable to overclaiming. The detector substantially improved expected behaviour by filtering unsupported answers. The stochastic uncertainty gate produced the strongest safety result by eliminating forbidden claim hits in the final benchmark.